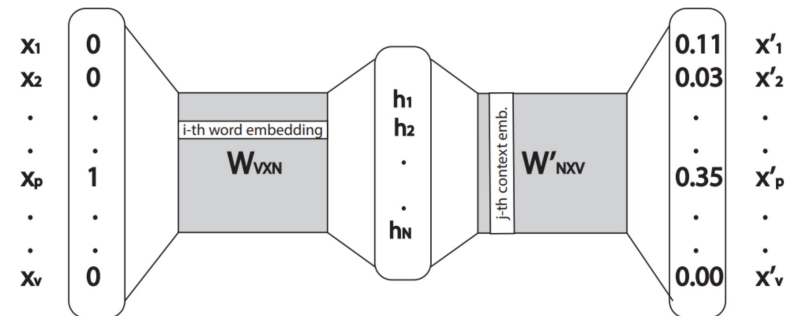
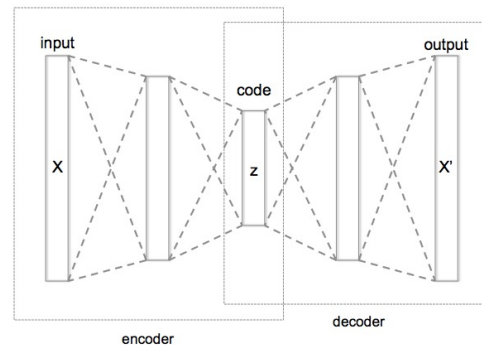


Embeddings

COMP6252: Deep Learning Technologies

University of Southampton UK



Xiaohao Cai

VLC / ECS / UoS

Content

- Dense to sparse representations
- Dimensionality reduction
- Word embeddings

Problem statement

> 171k words in
English vocabulary

- We need a vector representation of words

E.g. deep = [0, 0, 1, 0, 0, 0, 0, 0, 0, ..., 0]
 learning = [0, 0, 0, 0, 0, 1, 0, 0, 0, ..., 0]

One Hot Encoding

Issues:

- Very big vectors
- No notion of synonymy

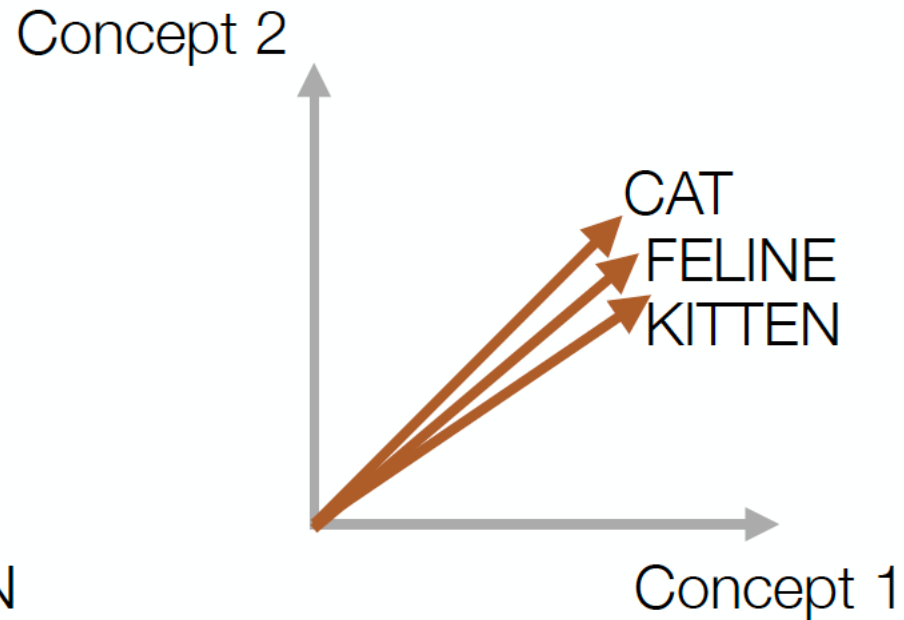
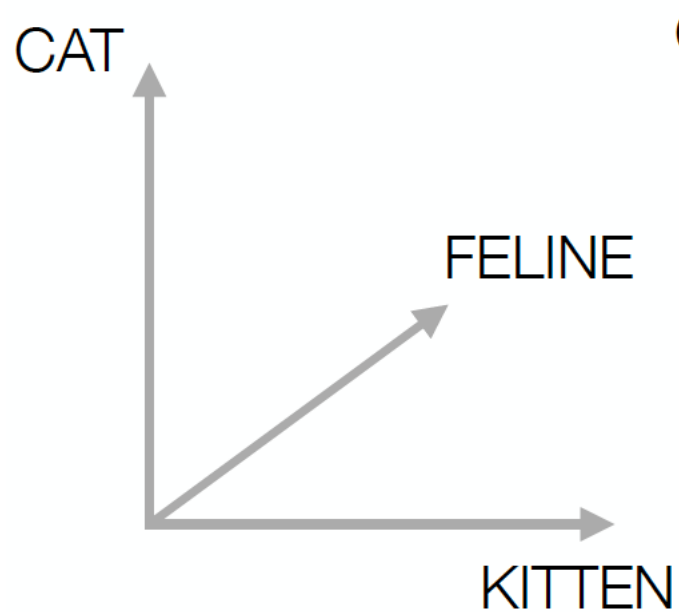
Dimensionality
reduction

- We prefer much lower dimensional vectors (far fewer weights)
- Vectors should capture similarity (e.g., cat -> kitten is closer than cat -> dog)

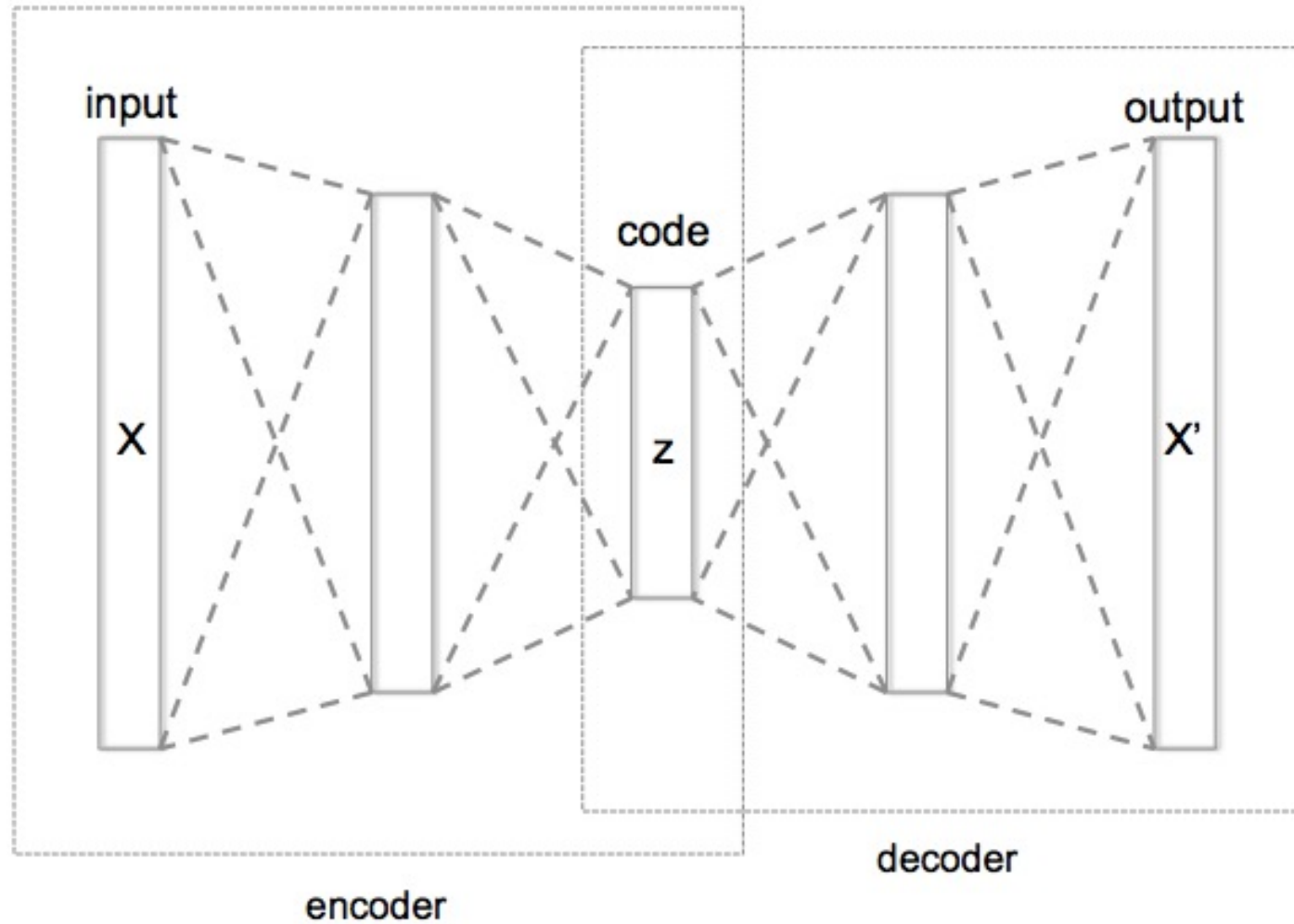
Learning Similarity

Distributional hypothesis:

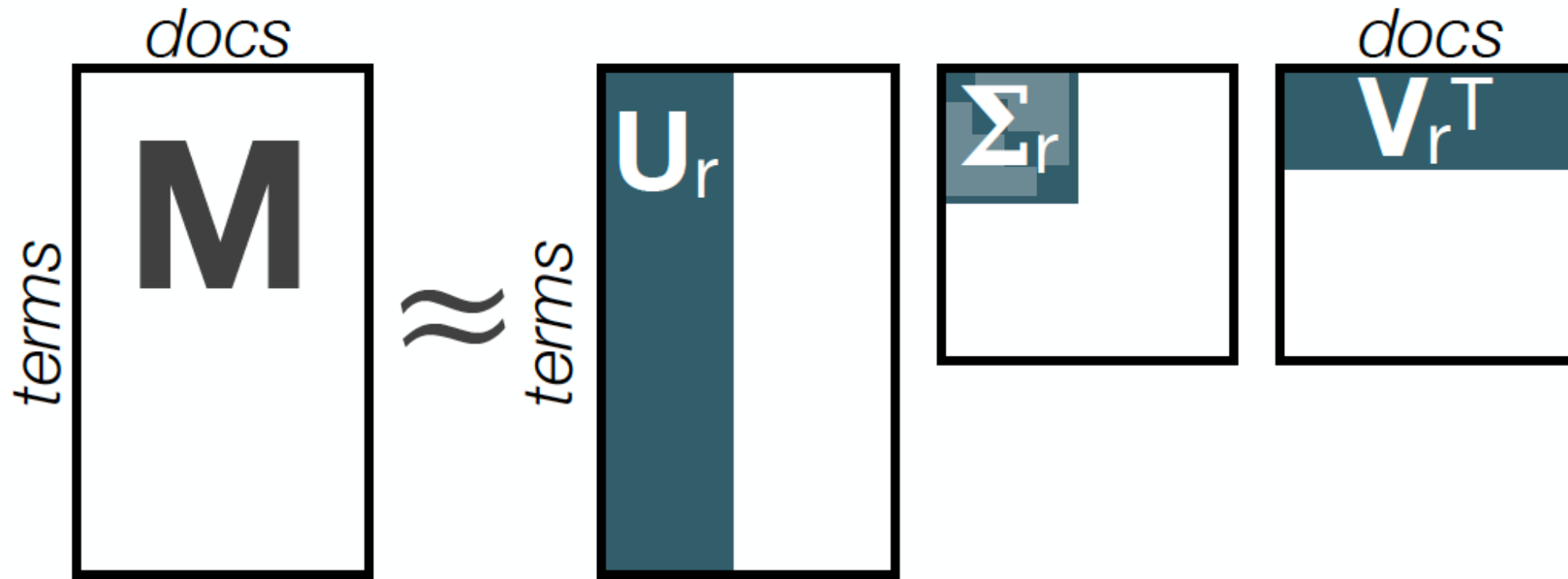
Words that are close in meaning will occur in similar pieces of text



Deep Autoencoder

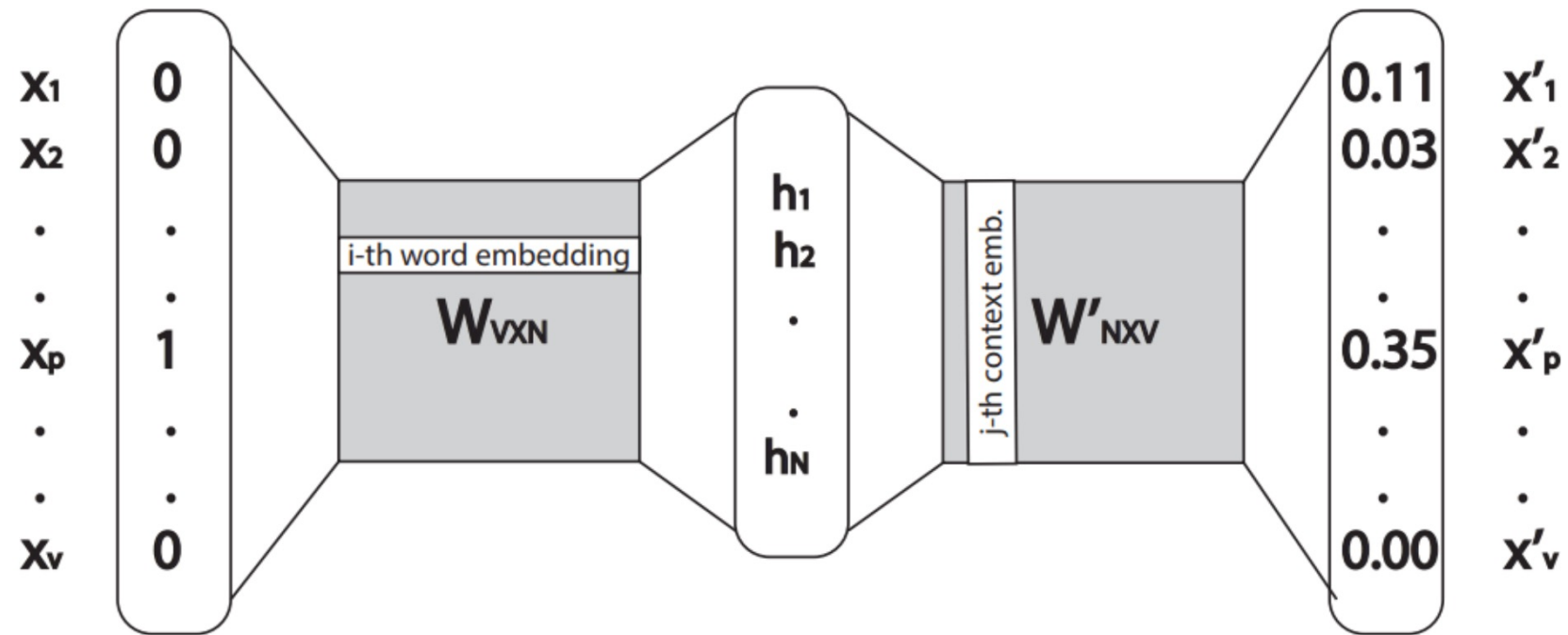


Latent Semantic Analysis



Truncated Singular Value Decomposition (T-SVD)

Word Embeddings - Word2vec



Word Embeddings

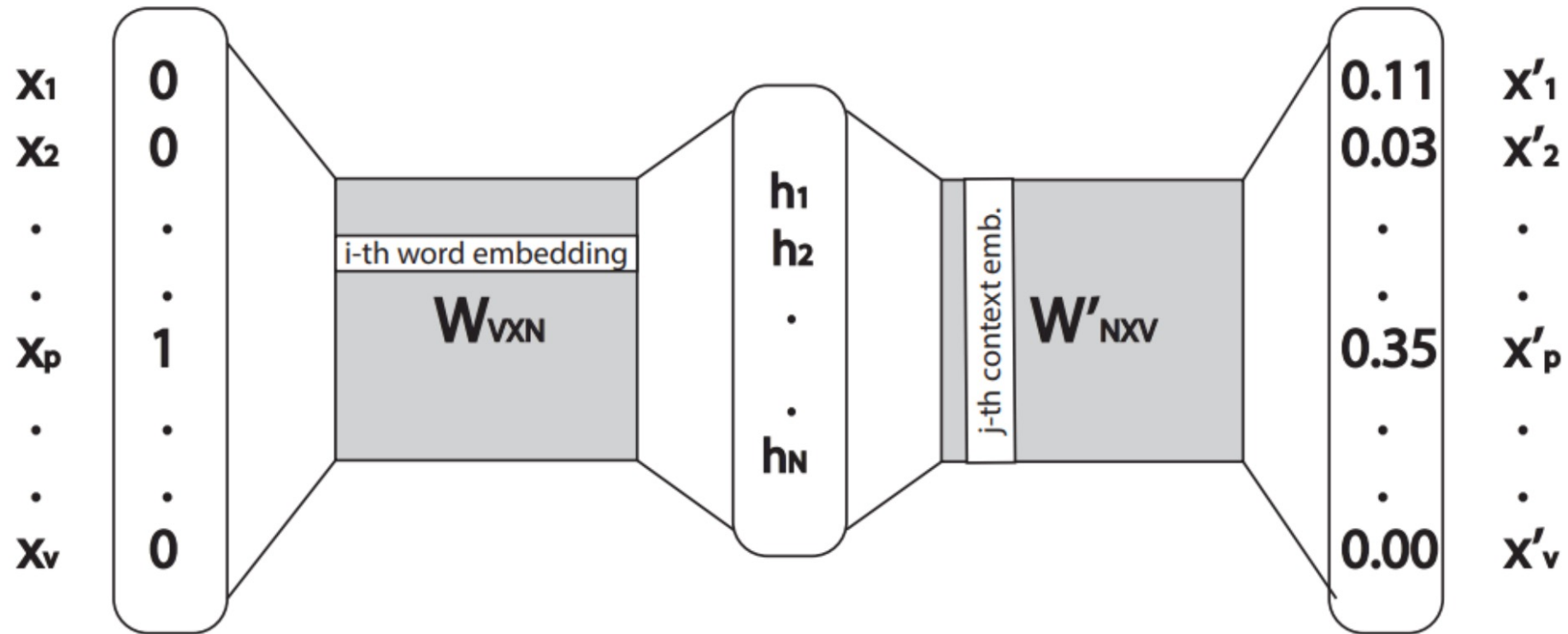
➤ Can we build a better vector representation of words?

- Lower dimensionality (but dense)
- Capturing synonymous words
- Capturing algebraic semantics



$$\text{word2vec}(\text{"Brother"}) - \text{word2vec}(\text{"Man"}) + \text{word2vec}(\text{"Woman"}) = \text{word2vec}(\text{"Sister"})$$

Word2vec



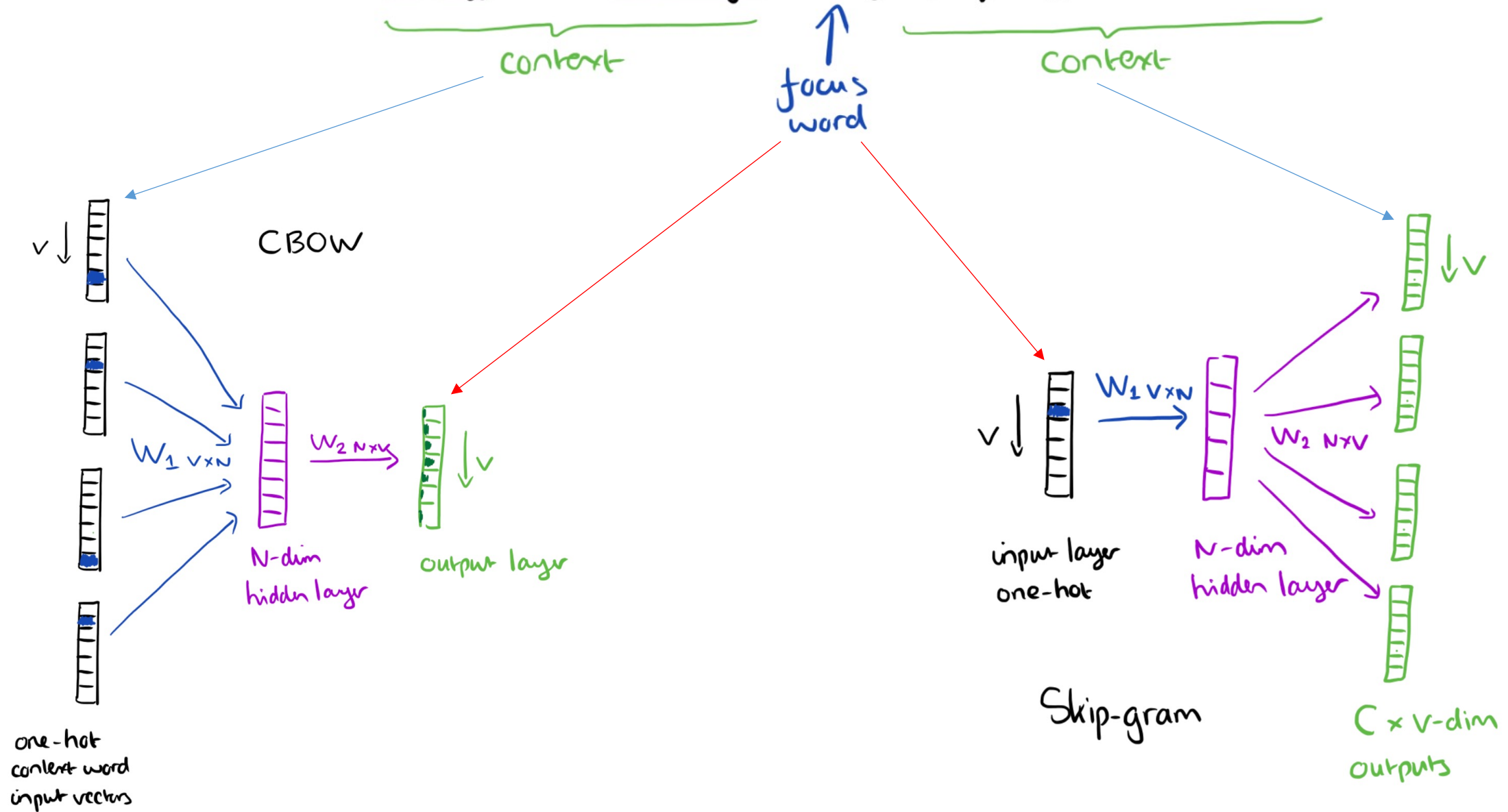
- Mikolov, Tomas; et al. (2013). "Efficient Estimation of Word Representations in Vector Space". arXiv:1301.3781
- Mikolov, Tomas; et al. (2013). Distributed representations of words and phrases and their compositionality. NeurIPS.

Word2vec

Source Text	Training Samples						
<table><tr><td>The</td><td>quick</td><td>brown</td></tr></table> fox jumps over the lazy dog. →	The	quick	brown	(the, quick) (the, brown)			
The	quick	brown					
<table><tr><td>The</td><td>quick</td><td>brown</td><td>fox</td></tr></table> jumps over the lazy dog. →	The	quick	brown	fox	(quick, the) (quick, brown) (quick, fox)		
The	quick	brown	fox				
<table><tr><td>The</td><td>quick</td><td>brown</td><td>fox</td><td>jumps</td></tr></table> over the lazy dog. →	The	quick	brown	fox	jumps	(brown, the) (brown, quick) (brown, fox) (brown, jumps)	
The	quick	brown	fox	jumps			
<table><tr><td>The</td><td>quick</td><td>brown</td><td>fox</td><td>jumps</td><td>over</td></tr></table> the lazy dog. →	The	quick	brown	fox	jumps	over	(fox, quick) (fox, brown) (fox, jumps) (fox, over)
The	quick	brown	fox	jumps	over		

Word2vec

...an efficient method for learning high quality distributed vector ...



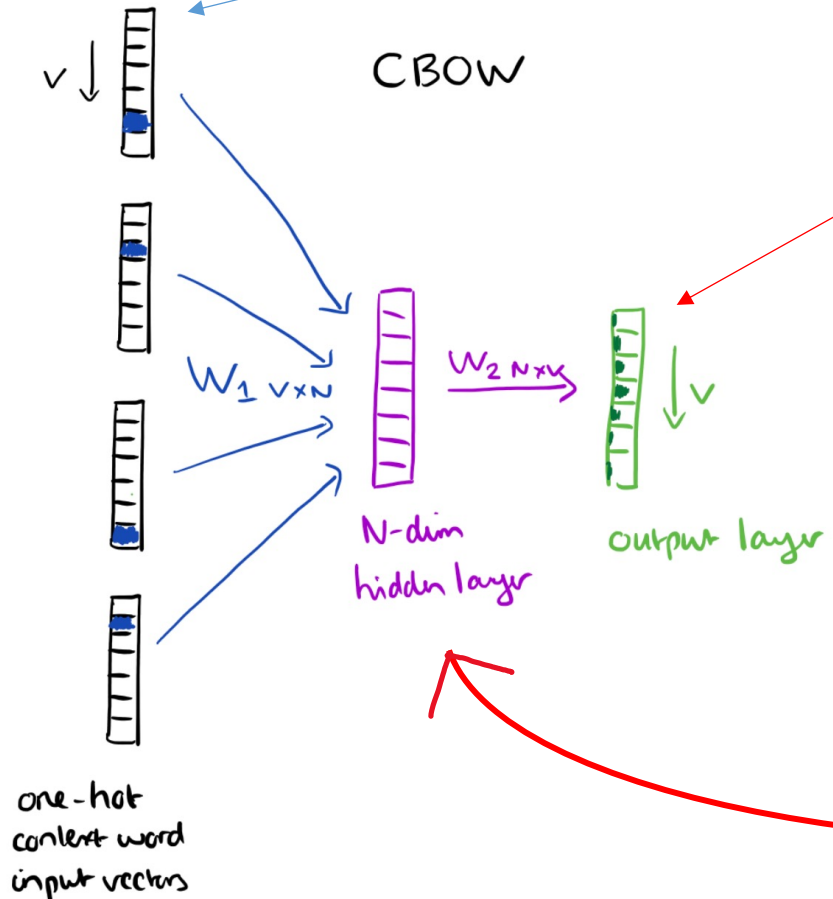
Word2vec

...an efficient method for learning high quality distributed vector ...

context

focus word

context



input $1 \times V$

W_1
 $V \times N$

hidden $1 \times N$

$$\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{bmatrix} = \begin{bmatrix} e & f & g & h \end{bmatrix}$$

W_1

Word2vec

Source Text	Training Samples			
<table><tr><td>The</td><td>quick</td><td>brown</td></tr></table> fox jumps over the lazy dog. ➡	The	quick	brown	(the, quick) (the, brown)
The	quick	brown		
The <table><tr><td>quick</td><td>brown</td><td>fox</td></tr></table> jumps over the lazy dog. ➡	quick	brown	fox	(quick, the) (quick, brown) (quick, fox)
quick	brown	fox		
The quick <table><tr><td>brown</td><td>fox</td><td>jumps</td></tr></table> over the lazy dog. ➡	brown	fox	jumps	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
brown	fox	jumps		
The quick brown <table><tr><td>fox</td><td>jumps</td><td>over</td></tr></table> the lazy dog. ➡	fox	jumps	over	(fox, quick) (fox, brown) (fox, jumps) (fox, over)
fox	jumps	over		

➤ “problems” with common words like e.g. “the”:

- When looking at word pairs, (“fox”, “the”) doesn’t tell us much about the meaning of “fox”. “the” appears in the context of pretty much every word. No notion of synonymy
- We will have many more samples of (“the”, ...) than we need to learn a good vector for “the”.

Word2vec

Source Text

Training Samples

The quick brown fox jumps over the lazy dog. →	(the, quick) (the, brown)
The quick brown fox jumps over the lazy dog. →	(quick, the) (quick, brown) (quick, fox)
The quick brown fox jumps over the lazy dog. →	(brown, the) (brown, quick) (brown, fox) (brown, jumps)
The quick brown fox jumps over the lazy dog. →	(fox, quick) (fox, brown) (fox, jumps) (fox, over)

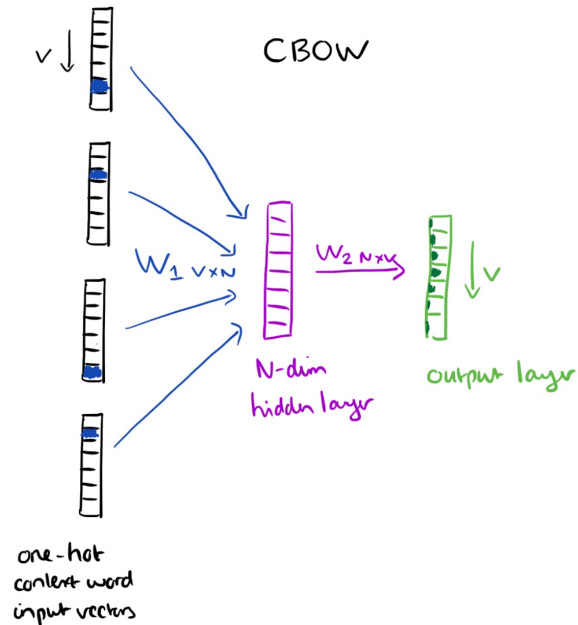
Improve training efficiency and quality:

- Subsampling
- Negative sampling

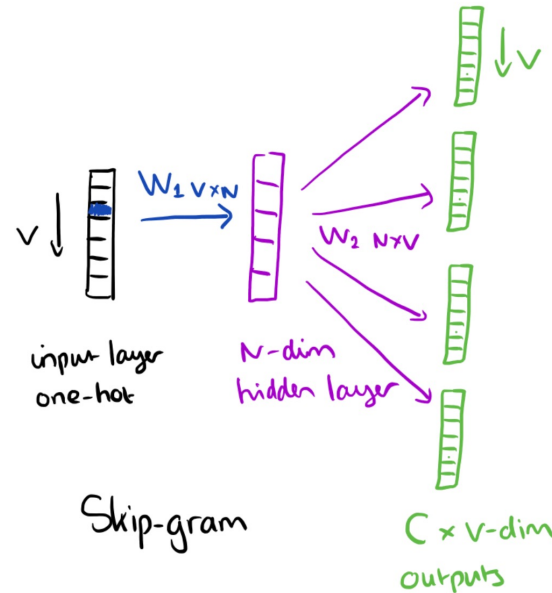
Word2vec codes:

- <https://radimrehurek.com/gensim/models/word2vec.html>
- <https://github.com/dav/word2vec>
- <https://www.geeksforgeeks.org/python-word-embedding-using-word2vec/>

Word2vec



Several times **faster to train** than the skip-gram, slightly better accuracy **for the frequent words**.



Works well with a **small amount of the training data**, represents well **even rare words or phrases**.

Limitations:

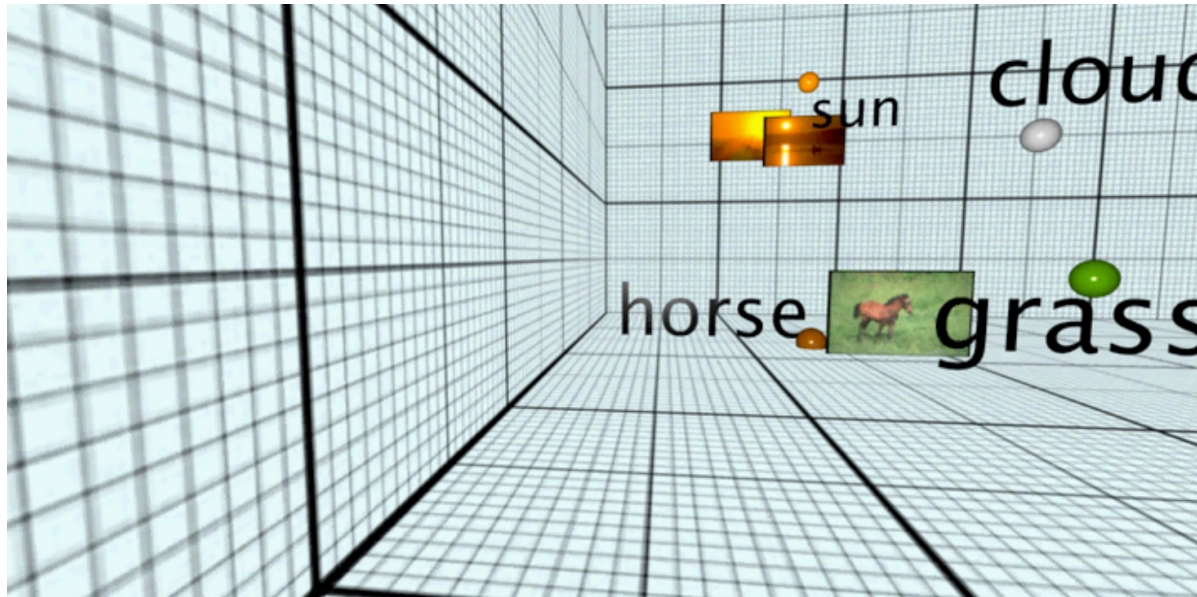
- Cannot deal with out of vocabulary (OOV) words
- More newer models

By 2022, the Word2vec approach was described as "**dated**", with **transformer models** being regarded as the **state of the art** in NLP

Image - Concept Embedding

➤ Basic idea:

Create a large multidimensional space in which **images, keywords (or other metadata) and visual information can be placed.**



Training:

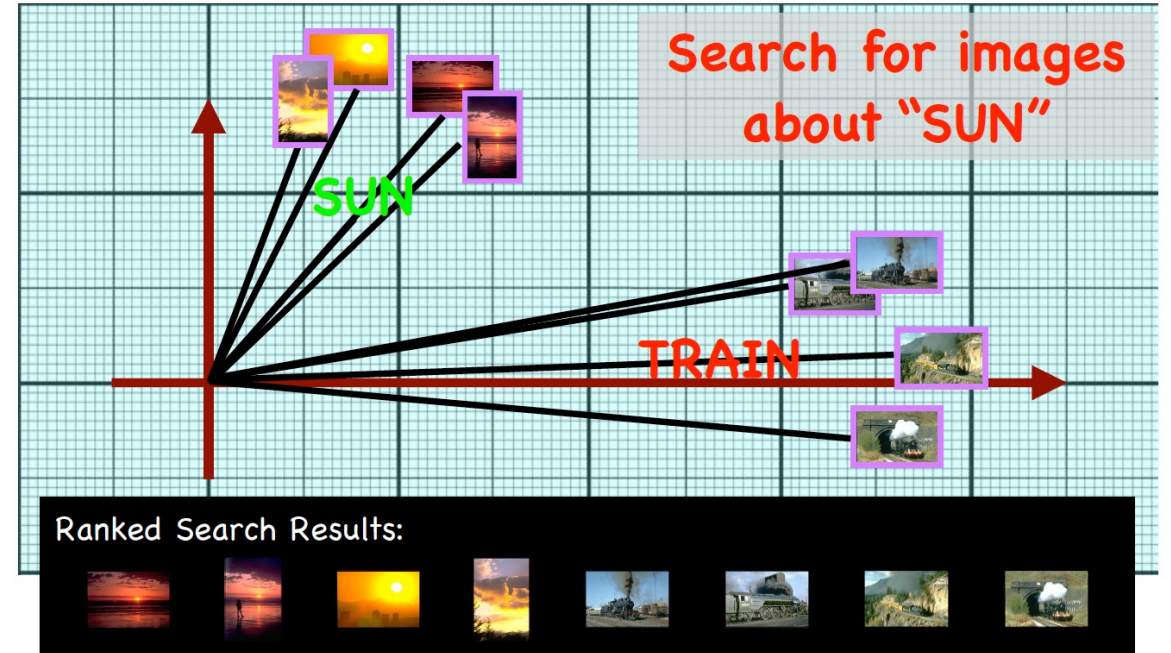
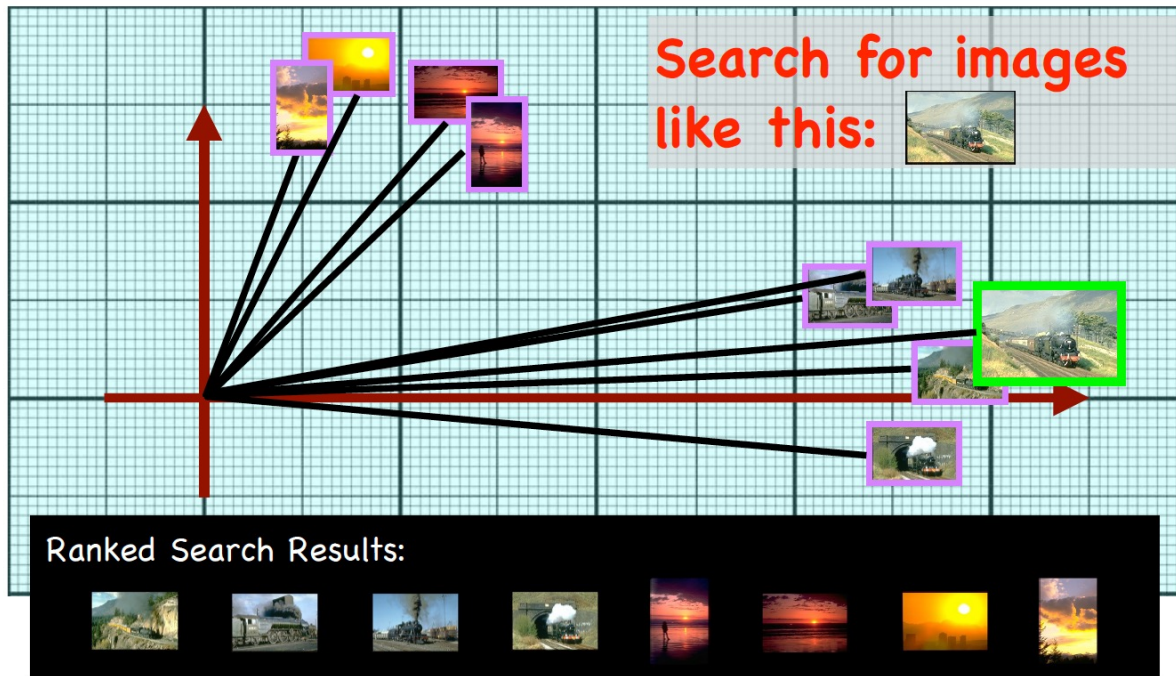
Learn how keywords are related to visual terms and images; and place related visual terms, images and keywords close-together within the space

Projection:

Unannotated images can be placed in the space based upon the visual terms they contain, such that they lie near keywords that describe them

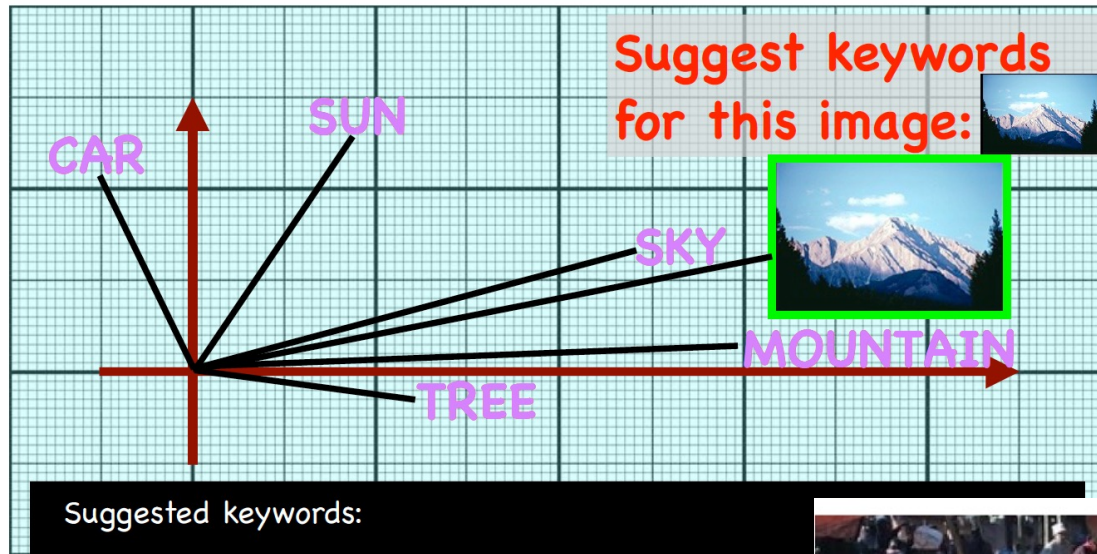
Image - Concept Embedding

➤ Searching by image



➤ Searching by Keyword

Image - Concept Embedding



Suggested keywords:

SKY MOUNTAIN TREE

- Suggesting keywords



- Image captioning

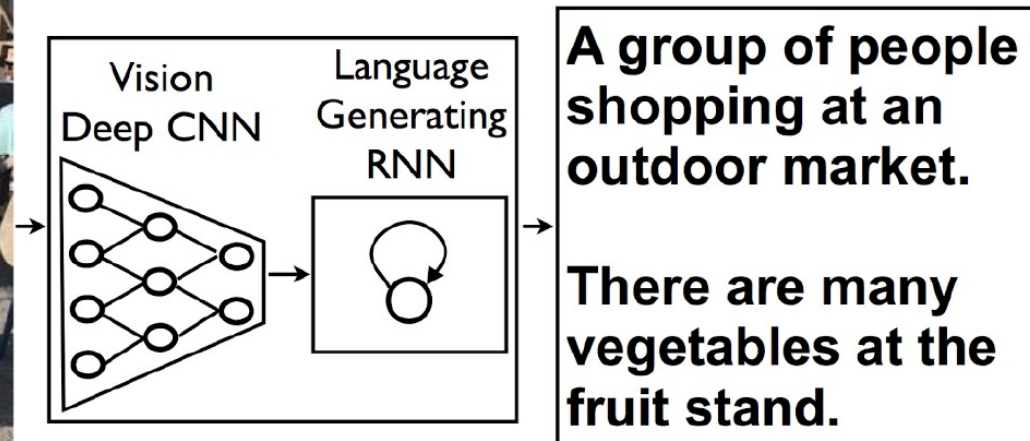
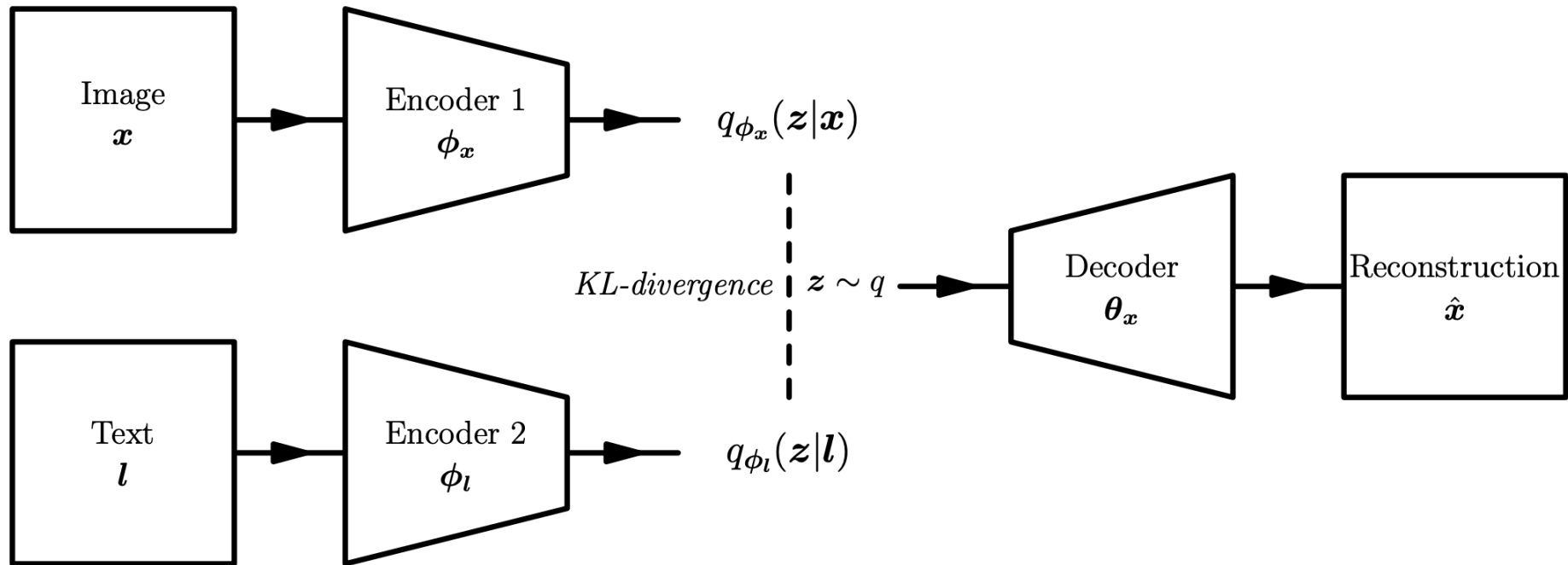


Image - Concept Embedding



➤ Probabilistic semantic embedding